

Subsection 1

Introduction

Summarizing data

- How to provide summary information about a large set of data?
 - ▶ Minimum value, maximum value, average value, median value
- Average value of a set of data is very useful in practice.
 - ▶ Average marks in a class exam
 - ▶ Run rate, batting average in cricket (and similar numbers in many sports)
- What does the average represent?
 - ▶ One number to represent a large set of numbers
 - ▶ Useful in comparisons and in so many scenarios
- Probability theory: expected value of a random variable is an important theoretical construct that represents the average value
 - ▶ It is used to express various types of summary information

Example: Casino math

Place bets on the sum seen when two die are rolled.

- Bets and returns
 - ▶ Under 7: get money back
 - ▶ Over 7: get money back
 - ▶ Equal to 7: get 4 times the money back

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Suppose you bet 1 unit of money.

Bet	Gain	Prob
Under 7	0	$5/12$
	-1	$7/12$
Over 7	0	$5/12$
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Equal 7	+3	$1/6$
	-1	$5/6$

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How did the casino decide on above returns? How to bet?

Expected value of a discrete random variable

Definition (Expected value)

Suppose X is a discrete random variable with range T_X and PMF f_X . The expected value of X , denoted $E[X]$, is defined as

$$E[X] = \sum_{t \in T_X} t f_X(t), = \sum_{t \in T_X} t \mathcal{P}(X=t)$$

assuming above sum exists.

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- Other names: mean of X , ~~average value~~ of X
- $E[X]$ may or may not belong to the range of X
- $E[X]$ has the same units as X

Examples: Easy summation given PMF

① $X \sim \text{Bernoulli}(p)$

$$X \in \{0, 1\}$$

$$E[X] = 0(1-p) + 1(p) = p = \mathcal{P}(X=1)$$

Examples: Easy summation given PMF

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$$E[X] = 0(1 - p) + 1(p) = p$$

- ② $X \sim \text{Uniform}\{1, 2, 3, 4, 5, 6\}$

$$E[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = 3.5$$

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- ③ Value of a lottery ticket (in Rs.) is $\left\{ \overset{1/1000}{200}, \overset{27/1000}{20}, \overset{972/1000}{0} \right\}$.

$$E[X] = 200 \cdot \frac{1}{1000} + 20 \cdot \frac{27}{1000} + 0 \cdot \frac{972}{1000} = \text{Rs. } 0.56$$

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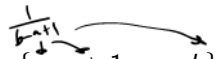
$$E[X] = 200 \cdot \frac{1}{1000} + 20 \cdot \frac{27}{1000} + 0 \cdot \frac{972}{1000} = \text{Rs. } 0.56$$

- ④ Change in temperature (in $^{\circ}\text{C}$) is $\left\{ \overset{1/5}{-2}, \overset{1/5}{-1}, \overset{1/2}{0}, \overset{1/10}{1} \right\}$.

$$E[X] = -2 \cdot \frac{1}{5} - 1 \cdot \frac{1}{5} + 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{10} = -0.5^{\circ}\text{C}$$

Example: $\text{Uniform}\{a, a + 1, \dots, b\}$

$X \sim \text{Uniform}\{a, a + 1, \dots, b\}$



$$E[X] = a \cdot \frac{1}{b - a + 1} + (a + 1) \cdot \frac{1}{b - a + 1} + \dots + b \cdot \frac{1}{b - a + 1}$$

Example: $\text{Uniform}\{a, a + 1, \dots, b\}$

$$X \sim \text{Uniform}\{a, a + 1, \dots, b\}$$

$$E[X] = a \cdot \frac{1}{b - a + 1} + (a + 1) \cdot \frac{1}{b - a + 1} + \dots + b \cdot \frac{1}{b - a + 1}$$

How to simplify sum?

$$\text{Identity: } a + (a + 1) + \dots + b = (b - a + 1) \frac{a+b}{2}$$

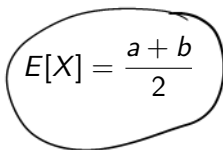
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How to simplify sum?

Identity: $a + (a + 1) + \dots + b = (b - a + 1) \frac{a+b}{2}$


$$E[X] = \frac{a + b}{2}$$

Examples: More involved summation

① $X \sim \text{Geometric}(p)$

$$X \in \{1, 2, \dots, k, \dots\} \quad (1-p)^{k-1}p$$

$$\begin{aligned} E[X] &= \sum_{t=1}^{\infty} t(1-p)^{t-1}p \\ &= p + 2(1-p)p + 3(1-p)^2p + \dots \end{aligned}$$

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① $X \sim \text{Geometric}(p)$

$$E[X] = \sum_{t=1}^{\infty} t(1-p)^{t-1}p$$

② $X \sim \text{Poisson}(\lambda)$

$$E[X] = \sum_{t=0}^{\infty} t e^{-\lambda} \frac{\lambda^t}{t!} \quad \text{P}(X=t)$$

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③ $X \sim \text{Binomial}(n, p)$

$$E[X] = \sum_{t=0}^n t \underbrace{\binom{n}{t} p^t (1-p)^{n-t}}_{\mathcal{P}(X=t)}$$

$\binom{n}{t} \leftrightarrow nC_t \leftrightarrow \frac{n!}{t!(n-t)!}$

How to simplify sum?

① Difference Equation (DE): $\underline{a_{t+1} - r a_t = b_t}$ ($r \neq 1$) $\xrightarrow{\text{another sequence}}$

$$\left(\sum_{t=1}^n a_t \right) = \frac{a_1 - r a_n}{1 - r} + \frac{1}{1 - r} \sum_{t=1}^{n-1} b_t$$

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$$\sum_{t=1}^n a_t = \frac{a_1 - r a_n}{1 - r} + \frac{1}{1 - r} \sum_{t=1}^{n-1} b_t$$

- ② Geometric Progression (GP): $a_{t+1} - r a_t = 0$ ($r \neq 1$)

$$\sum_{t=1}^n a_t = \frac{a_1 - r a_n}{1 - r} \quad \begin{matrix} |r| < 1 \\ \xrightarrow{n \rightarrow \infty} \end{matrix} \quad \frac{a_1}{1 - r}$$

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- ③ Exponential Function:

$$\sum_{t=0}^{\infty} e^{-\lambda} \frac{\lambda^t}{t!} = 1 \quad \Leftrightarrow \quad e^{\lambda} = \sum_{t=0}^{\infty} \frac{\lambda^t}{t!}$$

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- ③ Exponential Function:

$$\sum_{t=0}^{\infty} e^{-\lambda} \frac{\lambda^t}{t!} = 1$$

- ④ Binomial formula: $\sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = (a + b)^n$

$a = p, b = 1-p \Rightarrow \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = 1$ (RHS = 1)

Examples: More involved summation

- ① $X \sim \text{Geometric}(p)$ (DE: $a_1 = p$, $r = 1 - p$, $b_t = r^t p$)

$$E[X] = 1/p$$

$$E[X] = \sum_{t=1}^{\infty} \underbrace{t(1-p)^{t-1}}_{a_t} p = 1/p$$

$a_{t+1} = (t+1)(1-p)^t p$
 $a_{t+1} - (1-p)a_t = (1-p)^t p$

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- ② $X \sim \text{Poisson}(\lambda)$: $E[X] = \sum_{t=0}^{\infty} t e^{-\lambda} \frac{\lambda^t}{t!}$

$$E[X] = \lambda$$

$$\begin{aligned} E[X] &= e^{-\lambda} \left(0 \cdot 1 + 1 \cdot \frac{\lambda}{1!} + 2 \cdot \frac{\lambda^2}{2!} + 3 \cdot \frac{\lambda^3}{3!} + \dots \right) \\ &= \lambda e^{-\lambda} \left(1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots \right) = \lambda e^{-\lambda} e^{\lambda} = \lambda \end{aligned}$$

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- ① $X \sim \text{Geometric}(p)$ (DE: $a_1 = p$, $r = 1 - p$, $b_t = r^t p$)

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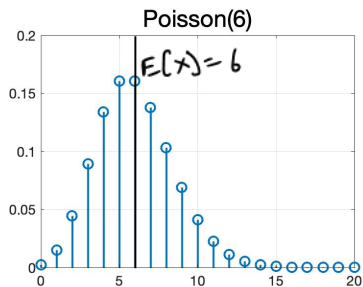
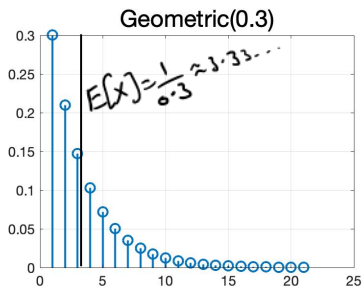
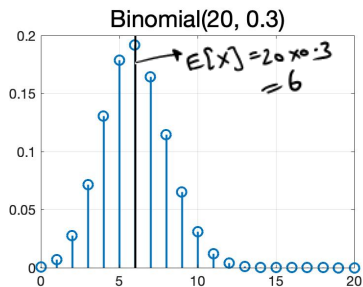
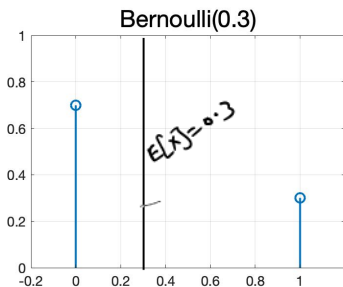
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- ③ $X \sim \text{Binomial}(n, p)$: $E[X] = \sum_{t=0}^n t \binom{n}{t} p^t (1-p)^{n-t}$

$$E[X] = np$$

$$\begin{aligned} E[X] &= \sum_{t=1}^n \frac{\cancel{t} n!}{\cancel{t}!(n-t)!} p^t (1-p)^{n-t} = np \sum_{t=1}^n \frac{(n-1)!}{(t-1)!(n-t)!} p^{t-1} (1-p)^{n-t} \\ &= np \underbrace{(p + (1-p))^{n-1}}_{=1} = \underbrace{(np)}_{\text{answer}} \end{aligned}$$

PMF and expected value



Subsection 2

Properties of expected value

Example: Casino math

Bet	Gain	Prob
Under 7	0	5/12
	-1	7/12
Over 7	0	5/12
	-1	7/12
Equal 7	+3	1/6
	-1	5/6

Suppose you bet 1 unit of money with probability p_1 each on “Under 7” or “Over 7” and probability $1 - 2p_1$ on “Equal to 7”. What is the expected gain?

$$\begin{aligned} & \overbrace{p_1 \cdot \frac{5}{12} \times 0 + p_1 \cdot \frac{7}{12} \times (-1)}^{\text{Under 7}} + \overbrace{p_1 \cdot \frac{5}{12} \times 0 + p_1 \cdot \frac{7}{12} \times (-1)}^{\text{Over 7}} + (1-2p_1) \cdot \frac{1}{6} \cdot 3 + (1-2p_1) \cdot \frac{5}{6} \cdot (-1) \\ &= \frac{-7p_1}{6} - (1-2p_1) \cdot \frac{2}{6} = \frac{-7p_1 - 2 + 4p_1}{6} = \frac{-(2+3p_1)}{6} \quad \underline{\underline{\text{Negative}}} \end{aligned}$$

Constant random variable and positive random variable

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- ① Consider a constant c as a random variable X with $P(X = c) = 1$.

$$E[c] = c \cdot 1 = c$$

- ② Suppose X takes only non-negative values, i.e. $P(X \geq 0) = 1$. Then,

$$E[X] \geq 0$$

Expected value of a function of random variables

Theorem (Expected value of a function)

Suppose $X_1, \dots, X_n$ have joint PMF $f_{X_1 \dots X_n}$ with range of X_i denoted T_{X_i} . Let $g : T_{X_1} \times \dots \times T_{X_n} \rightarrow \mathbb{R}$ be a function, and let $Y = \underline{\underline{g(X_1, \dots, X_n)}}$ have range T_Y and PMF f_Y . Then,

$$E[g(X_1, \dots, X_n)] = \sum_{t \in T_Y} \overbrace{t}^{E[Y]} f_Y(t) = \sum_{t_i \in T_{X_i}} g(t_1, \dots, t_n) f_{X_1 \dots X_n}(t_1, \dots, t_n).$$

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- We have seen how to find f_Y , PMF of a function of multiple random variables
- The above theorem says: To find $E[Y]$, you do not always need f_Y . The joint PMF of $X_1, \dots, X_n$ can be used directly.
- Simple sounding property with far-reaching consequences!

Examples for illustration

$$\textcircled{1} \quad X \sim \left\{ \overset{1/5}{-2}, \overset{1/5}{-1}, \overset{1/5}{0}, \overset{1/5}{1}, \overset{1/5}{2} \right\}, \quad g(X) = X^2 \sim \left\{ \overset{1/5}{0}, \overset{2/5}{1}, \overset{2/5}{4} \right\}$$

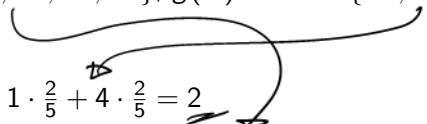
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$$E[g(X)] = 0 \cdot \frac{1}{5} + 1 \cdot \frac{2}{5} + 4 \cdot \frac{2}{5} = 2$$

Examples for illustration

① $X \sim \{-2, -1, 0, 1, 2\}$, $g(X) = X^2 \sim \{0, 1, 4\}$



$$E[g(X)] = 0 \cdot \frac{1}{5} + 1 \cdot \frac{2}{5} + 4 \cdot \frac{2}{5} = 2$$

$$E[g(X)] = (-2)^2 \cdot \frac{1}{5} + (-1)^2 \cdot \frac{1}{5} + (0)^2 \cdot \frac{1}{5} + (1)^2 \cdot \frac{1}{5} + (2)^2 \cdot \frac{1}{5} = 2$$

$g(x)$ ↓ $P(X=-2)$

$\sum_t t^2 P(X=t)$

Examples for illustration

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$$\textcircled{2} (X, Y) \sim \text{Uniform}\left\{ (0, 0), (1, 0), (0, 1), (1, 1), (-1, 1), (1, -1) \right\}$$

$$g(X, Y) = X^2 + XY + Y^2 \sim \left\{ 0, 1, 3 \right\}$$

x	y	$g(x, y)$	P_{xy}
0	0	0	$\frac{1}{6}$
1	0	1	$\frac{1}{6}$
0	1	1	$\frac{1}{6}$
1	1	3	$\frac{1}{6}$
-1	1	1	$\frac{1}{6}$
1	-1	1	$\frac{1}{6}$

Examples for illustration

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$$E[g(X)] = 0 \cdot \frac{1}{5} + 1 \cdot \frac{2}{5} + 4 \cdot \frac{2}{5} = 2$$

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$$g(X, Y) = X^2 + XY + Y^2 \sim \left\{ \overset{1/6}{0}, \overset{4/6}{1}, \overset{1/6}{3} \right\}$$

$$E[g(X, Y)] = 0 \cdot \frac{1}{6} + 1 \cdot \frac{4}{6} + 3 \cdot \frac{1}{6} = 7/6$$

Examples for illustration

$$\textcircled{1} X \sim \left\{ -2, -1, 0, 1, 2 \right\}, g(X) = X^2 \sim \left\{ 0, 1, 4 \right\}$$

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$$E[g(X, Y)] = 0 \cdot \frac{1}{6} + 1 \cdot \frac{4}{6} + 3 \cdot \frac{1}{6} = 7/6$$

$$E[g(X, Y)] = 0 \cdot \frac{1}{6} + (1) \cdot \frac{1}{6} + (1) \cdot \frac{1}{6} + (3) \cdot \frac{1}{6} + (1) \cdot \frac{1}{6} + (1) \cdot \frac{1}{6} = 7/6$$

Handwritten annotations:
Below the first 0: $g(0,0)$
Below the first 1: $g(1,0)$
Below the first 1: $g(0,1)$
Below the 3: $g(1,1)$
Below the second 1: $g(-1,1)$
Below the second 1: $g(1,-1)$

Linearity of expected value

- ① $E[cX]$ = $cE[X]$ for a random variable X and a constant c

Proof:

$$E[cX] = \sum_{t \in T_X} ct f_X(t) = c \sum_{t \in T_X} t f_X(t) = cE[X]$$

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- ② $E[X + Y] = E[X] + E[Y]$ for any two random variables X, Y

Proof:

$$\begin{aligned} E[X + Y] &= \sum_{t_1 \in T_X, t_2 \in T_Y} (t_1 + t_2) f_{XY}(t_1, t_2) \\ &= \sum_{t_1 \in T_X, t_2 \in T_Y} t_1 f_{XY}(t_1, t_2) + \sum_{t_1 \in T_X, t_2 \in T_Y} t_2 f_{XY}(t_1, t_2) \\ &= E[X] + E[Y] \end{aligned}$$

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- ② $E[X + Y] = E[X] + E[Y]$ for any two random variables X, Y

Proof:

$$\begin{aligned} E[X + Y] &= \sum_{t_1 \in T_X, t_2 \in T_Y} (t_1 + t_2) f_{XY}(t_1, t_2) \\ &= \sum_{t_1 \in T_X, t_2 \in T_Y} t_1 f_{XY}(t_1, t_2) + \sum_{t_1 \in T_X, t_2 \in T_Y} t_2 f_{XY}(t_1, t_2) \\ &= E[X] + E[Y] \end{aligned}$$

$$E[aX + bY] = aE[X] + bE[Y]$$

One of the most useful properties of expected value!

Illustrative examples

$$\textcircled{1} (X, Y) \sim \text{Uniform}\{(0, 0), (1, 0), (0, 1), (1, 1), (-1, 1), (1, -1)\}$$

$$g(X, Y) = X^2 + XY + Y^2 \sim \left\{ \overset{1/6}{0}, \overset{4/6}{1}, \overset{1/6}{3} \right\}$$

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$$(E[X^2] = 4/6, E[XY] = -1/6, E[Y^2] = 4/6)$$

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$$g(X, Y) = X^2 + XY + Y^2 \sim \begin{matrix} 1/6 & 4/6 & 1/6 \\ 0 & 1 & 3 \end{matrix}$$

$$E[g(X, Y)] = 7/6$$

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$$E[X + Y] = E[X] + E[Y] = \sum_{t \in T_X} t f_X(t) + \sum_{t \in T_Y} t f_Y(t)$$

Note: Expected value of the form $E[g(X) + h(Y)]$ can be computed with marginal PMFs, and it does not depend on the joint PMF.

Problem: Throw a fair die twice

What is the expected value of the sum of the two numbers seen?

X : first number Y : second number

$$\begin{aligned} E[X+Y] &= E[X] + E[Y] \\ &= 3.5 + 3.5 = 7 \end{aligned}$$

Problem: Expected value of Binomial(n, p)

Suppose $Y \sim \text{Binomial}(n, p)$.

$$E[Y] = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k} = np$$

- Such summations may be difficult to simplify

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- Alternative method using linearity of expected value

Let $X_1, \dots, X_n$ be i.i.d. Bernoulli(p). Then, $E[X_i] = p$ and

$$Y = X_1 + \dots + X_n \sim \text{Binomial}(n, p)$$

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$$Y = X_1 + \dots + X_n \sim \text{Binomial}(n, p)$$

So, $E[Y] = E[X_1] + \dots + E[X_n] = np$

Zero-mean random variable

A random variable X with $E[X] = 0$ is said to be a zero-mean random variable.

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Translation of a random variable

$X + c$, where c is a constant, is a “translated” version of X .

- Range of $X + c$ is $\{t + c : t \in T_X\}$, which is a translated version of T_X , the range of X
- $P(X + c = t + c) = P(X = t)$ and the PMF is “translated” as well

Zero-mean random variable

A random variable X with $E[X] = 0$ is said to be a zero-mean random variable.

Translation of a random variable

$$E[X+c] = E[X] + c$$

$X + c$, where c is a constant, is a “translated” version of X .

- Range of $X + c$ is $\{t + c : t \in T_X\}$, which is a translated version of T_X , the range of X
- $P(X + c = t + c) = P(X = t)$ and the PMF is “translated” as well

Translation by expected value

Important: $E[X]$ is a constant.

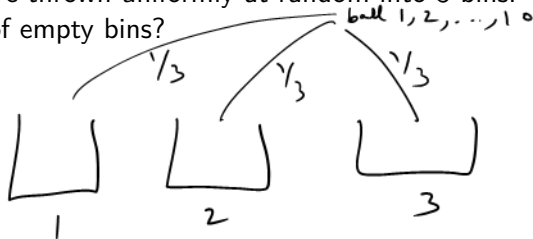
$Y = X - E[X]$ is a translated version of X and $E[Y] = 0$.

So, $X - E[X]$ is a zero-mean random variable.

“centering”

Problem: Balls on bins

Suppose 10 balls are thrown uniformly at random into 3 bins. What is the expected number of empty bins?



$$X_i = \begin{cases} 1, & \text{if bin } i \text{ is empty} \\ 0, & \text{otherwise} \end{cases} \quad i = 1, 2, 3$$

$$P(X_i = 1) = \frac{2^{10}}{3^{10}} = \left(\frac{2}{3}\right)^{10}, \quad P(X_i = 0) = 1 - \left(\frac{2}{3}\right)^{10}$$

$$Y = \# \text{ empty bins} = X_1 + X_2 + X_3 \quad \text{"Finding } f_Y \text{ is a little hard"}$$
$$E[Y] = E[X_1] + E[X_2] + E[X_3] = 3 \cdot \left(\frac{2}{3}\right)^{10}$$

Subsection 3

Variance

Motivation

How good is expected value in describing a random variable?

- Let $X \sim \{10\}$, $Y \sim \{9^{1/2}, 11^{1/2}\}$, $Z \sim \{0^{1/2}, 20^{1/2}\}$.

$$E[X] = E[Y] = E[Z] = 10$$

However, the three random variables are quite different!

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However, the three random variables are quite different!

Center and spread

- Expected value: represents “center” of a random variable
- We need some indicator of “spread” about the expected value

Variance and standard deviation

Definition (Variance and standard deviation)

The variance of a random variable X , denoted $\text{Var}(X)$, is defined as

$$\text{Var}(X) = E[(X - \overbrace{E[X]}^{\text{some constant}})^2].$$

function of X

The standard deviation of X , denoted $\text{SD}(X)$, is defined as

$$\text{SD}(X) = +\sqrt{\text{Var}(X)}.$$

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The standard deviation of X , denoted $\text{SD}(X)$, is defined as

$$\text{SD}(X) = +\sqrt{\text{Var}(X)}.$$

- Variance is expected value of the random variable $(X - E[X])^2$.
 - ▶ $\text{Var}(X) = \sum_{t \in T_X} (t - E[X])^2 P(X = t)$
 - ▶ Variance is non-negative, and standard deviation is well-defined.
 - ▶ Units of $\text{SD}(X)$ are same as units of X .
- Intuitively, the more the “spread” in T_X , the more will be $\text{Var}(X)$

Example

$$X \sim \{ \overset{1}{10} \}, Y \sim \{ \overset{1/2}{9}, \overset{1/2}{11} \}, Z \sim \{ \overset{1/2}{0}, \overset{1/2}{20} \}$$

$$E[X] = E[Y] = E[Z] = 10$$

Variances and standard deviations are different!

Example

$$X \sim \{10\}, Y \sim \{9^{1/2}, 11^{1/2}\}, Z \sim \{0^{1/2}, 20^{1/2}\}$$

$$E[X] = E[Y] = E[Z] = 10$$

Variances and standard deviations are different!

$$\text{Var}(X) = (10 - 10)^2 \cdot 1 = 0, \text{SD}(X) = 0$$

Handwritten annotations:
An arrow points from $E[X]$ to the first 10 in the variance formula.
An arrow points from $E[X]$ to the second 10 in the variance formula.

Example

$$X \sim \{10\}, Y \sim \{9, 11\}, Z \sim \{0, 20\}$$

$$E[X] = E[Y] = E[Z] = 10$$

Variances and standard deviations are different!

$$\text{Var}(X) = (10 - 10)^2 \cdot 1 = 0, \text{SD}(X) = 0$$

$$\text{Var}(Y) = (9 - 10)^2 \cdot \frac{1}{2} + (11 - 10)^2 \cdot \frac{1}{2} = 1, \text{SD}(Y) = 1$$

Handwritten annotations:
- $t \in T_Y$ points to the 9 in the first term.
- $E[Y]$ points to the 10 in the first term.
- $P(Y=t)$ points to the $\frac{1}{2}$ in the first term.

Example

$$X \sim \{10\}, Y \sim \{9^{1/2}, 11^{1/2}\}, Z \sim \{0^{1/2}, 20^{1/2}\}$$

$$E[X] = E[Y] = E[Z] = 10$$

Variances and standard deviations are different!

$$\text{Var}(X) = (10 - 10)^2 \cdot 1 = 0, \text{SD}(X) = 0$$

$$\text{Var}(Y) = (9 - 10)^2 \cdot \frac{1}{2} + (11 - 10)^2 \cdot \frac{1}{2} = 1, \text{SD}(X) = 1$$

$$\text{Var}(Z) = (0 - 10)^2 \cdot \frac{1}{2} + (20 - 10)^2 \cdot \frac{1}{2} = 100, \text{SD}(X) = 10$$

Example: Throw a die

$X \sim \text{Uniform}\{1, 2, 3, 4, 5, 6\}$

$$E[X] = 3.5$$

Example: Throw a die

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$$E[X] = 3.5$$

$$\begin{aligned}\text{Var}(X) &= (1 - 3.5)^2 \cdot \frac{1}{6} + (2 - 3.5)^2 \cdot \frac{1}{6} + (3 - 3.5)^2 \cdot \frac{1}{6} \\ &\quad + (4 - 3.5)^2 \cdot \frac{1}{6} + (5 - 3.5)^2 \cdot \frac{1}{6} + (6 - 3.5)^2 \cdot \frac{1}{6} \\ &= \frac{35}{12} = 2.916 \dots\end{aligned}$$

$$\text{SD}(X) = 1.7078 \dots$$


$$\sqrt{\frac{35}{12}}$$

Example: Throw a die

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$$\text{SD}(X) = 1.7078\dots$$

Given PMF of X with a small range, variance and standard deviation can be readily computed.

Properties: Scaling and translation

Let X be a random variable. Let a be a constant real number.

① $\text{Var}(aX) = a^2 \text{Var}(X)$

② $\text{SD}(aX) = |a| \text{SD}(X)$

③ $\text{Var}(X + a) = \text{Var}(X)$

④ $\text{SD}(X + a) = \text{SD}(X)$

Pf.
$$\begin{aligned} E[(X+a - \underbrace{E[X+a]}_a)^2] \\ = E[(X + \cancel{X} - E[X] - a)^2] \end{aligned}$$

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- ③ $\text{Var}(X + a) = \text{Var}(X)$
- ④ $\text{SD}(X + a) = \text{SD}(X)$
 - Contrast with similar properties for $E[X]$

Alternative formula for variance

Theorem

The variance of a random variable X is given by

$$\text{Var}(X) = E[X^2] - E[X]^2.$$

$$E[X^2] = \sum_{t \in T_X} t^2 P(X=t)$$

$$E[X]^2 = \left(\sum_{t \in T_X} t P(X=t) \right)^2$$

Alternative formula for variance

Theorem

The variance of a random variable X is given by

$$\text{Var}(X) = E[X^2] - E[X]^2.$$

Proof:

- $\text{Var}(X) = E[(X - E[X])^2] = E[X^2 - 2E[X]X + E[X]^2]$
- By linearity of expected value,

$$\text{Var}(X) = E[X^2] - 2E[E[X]X] + E[E[X]^2]$$

- Since $E[X]$ is a constant,

$$\text{Var}(X) = E[X^2] - 2E[X]E[X] + E[X]^2 = E[X^2] - E[X]^2$$

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$$\text{Var}(X) = E[X^2] - 2E[X]E[X] + E[X]^2 = E[X^2] - E[X]^2$$

$E[X]$, $E[X^2]$: first and second moment, $\text{Var}(X)$: second central moment

Sum and product of independent random variables

For any two random variables X and Y (independent or dependent),

$$E[X + Y] = E[X] + E[Y].$$

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Suppose X and Y are **independent** random variables. Then, more is true.

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Suppose X and Y are **independent** random variables. Then, more is true.

- ① $E[XY] = E[X]E[Y]$
- ② $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

Proof

$$\begin{aligned}
 E[XY] &= \sum_{\substack{t_1 \in T_X \\ t_2 \in T_Y}} t_1 t_2 \underbrace{P(X=t_1, Y=t_2)}_{X \& Y \text{ indep: } = P(X=t_1)P(Y=t_2)} \\
 &= \sum_{\substack{t_1 \in T_X \\ \text{over } t_2}} \left(\sum_{t_2 \in T_Y} t_1 t_2 P(X=t_1) P(Y=t_2) \right) \xrightarrow{\text{factoring is crucial}} \\
 &= \sum_{t_1 \in T_X} \left(t_1 P(X=t_1) \left(\sum_{t_2 \in T_Y} t_2 P(Y=t_2) \right) \right) \\
 &= E[Y] \left(\sum_{t_1 \in T_X} t_1 P(X=t_1) \right) \\
 &= E[X] E[Y] \rightarrow \text{Not always true if } X \& Y \text{ are dependent}
 \end{aligned}$$

Sum and product of independent random variables

For any two random variables X and Y (independent or dependent),

$$E[X + Y] = E[X] + E[Y].$$

Suppose X and Y are **independent** random variables. Then, more is true.

① $E[XY] = E[X]E[Y]$ 

② $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ Pf: Exercise.

Example: Sum of two dice (independent)

What is the variance of the sum of two dice?

X : first die Y : second die

X & Y : independent

$$\begin{aligned}\text{Var}(X+Y) &= \text{Var}(X) + \text{Var}(Y) \\ &= 2 \sqrt{\frac{35}{12}} = \sqrt{\frac{35}{3}}\end{aligned}$$

Variance of common distributions

Distribution	Expected value	Variance
Bernoulli(p)	p	$p(1-p)$
Binomial(n, p)	np	$np(1-p)$
Geometric(p)	$1/p$	$(1-p)/p^2$
Poisson(λ)	λ	λ
Uniform $\{1, \dots, n\}$	$(n+1)/2$	$(n^2-1)/12$

① $X \sim \{0, 1\}$ $E[X] = p$ $Var(X) = p - p^2$
 $E[X^2] = 1 \cdot p = p \rightarrow p(1-p)$

② $X \sim \text{Binomial}(n, p) \rightarrow E[X] = np$
 $X = X_1 + \dots + X_n \rightarrow Var(X) = Var(X_1) + \dots + Var(X_n)$
 $X_i \sim \text{iid Bernoulli}(p) \rightarrow = np(1-p)$ (more involved)
 $E[X^2] = \sum_{k=0}^n k^2 \binom{n}{k} p^k (1-p)^{n-k} = \underbrace{np(1-p)}_{Var(X)} + \underbrace{(np)^2}_{E[X]^2}$

Working

Standardised random variables

Definition

A random variable X is said to be standardised if

$$E[X] = 0, \text{Var}(X) = 1.$$

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Theorem

Let X be a random variable. Then,

$$Y = \frac{X - E[X]}{SD(X)}$$

is a standardised random variable.

$$E[Y] = \frac{1}{SD(X)} (E[X] - E[X]) = 0 \quad \text{Var}(Y) = \text{Var}\left(\frac{X}{\underbrace{SD(X)}_{\text{constant}}}\right) = \frac{\text{Var}(X)}{SD(X)^2} = 1$$

→ PMF of Y is "very similar" to PMF of X .

Existence of expected value and variance

- There are random variables s.t. $E[X]$ goes to ∞ (or $-\infty$)
 - ▶ $X \sim \{1^{1/2}, 2^{1/4}, 4^{1/8}, \dots, 2^{n-1}, \dots\}$
 - ▶ $E[X^2]$ will go to ∞ in this case and $\text{Var}(X)$ is ill-defined
- There are random variables s.t. $E[X]$ is not well-defined
 - ▶ $X \sim \{1^{1/2}, -2^{1/4}, 4^{1/8}, \dots, (-2)^{n-1}, \dots\}$
 - ▶ $E[X^2]$ will go to ∞ in this case and $\text{Var}(X)$ is ill-defined
- There are random variables s.t. $E[X]$ is finite, but $E[X^2]$ goes to ∞
 - ▶ $\text{Var}(X)$ goes to ∞ too $X \in \{1, 2, 3, \dots\} \quad P(X=k) \propto \frac{1}{k^3}$
- In this course, we will generally consider well-behaved random variables with finite mean and variance

Subsection 4

Covariance and correlation

Motivation

Consider the following two joint PMFs for two random variables X and Y .

f_{XY}	$x=0$	$x=1$	f_Y
$y=0$	$1/4$	$1/4$	$1/2$
$y=1$	$1/4$	$1/4$	$1/2$
f_X	$1/2$	$1/2$	

f_{XY}	0	1	f_Y
0	0	$1/2$	$1/2$
1	$1/2$	0	$1/2$
f_X	$1/2$	$1/2$	

Motivation

Consider the following two joint PMFs for two random variables X and Y .

f_{XY}	0	1
0	1/4 1/4	1/4 1/4
1	1/4 1/4	1/4 1/4

f_{XY}	0	1
0	0	1/2
1	1/2	0

- Same marginal PMFs in both cases. So, mean and variance are the same for X and Y .
- However, the two cases are very different
 - ▶ X and Y are independent in one case
 - ▶ Value of X determines the value of Y in the other

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How to summarize relationship between random variables?

Covariance

Definition (Covariance)

Suppose X and Y are random variables on the same probability space. The covariance of X and Y , denoted $\text{Cov}(X, Y)$, is defined as

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])].$$

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- $\text{Cov}(X, Y)$ is positive
 - ▶ $(X - E[X])(Y - E[Y])$ tends to be positive
 - ▶ When X is above/below its average, Y tends to be correspondingly above/below its average

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- $\text{Cov}(X, Y)$ is negative
 - ▶ $(X - E[X])(Y - E[Y])$ tends to be negative
 - ▶ When X is above/below its average, Y tends to be correspondingly below/above its average
- $\text{Cov}(X, Y) = 0$
 - ▶ X and Y are said to be “uncorrelated”

Examples: positive and negative covariance

- 1 X : height of a person, Y : weight of a person
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- ③ X : Runs in an IPL over, Y : Wickets in the same over
 - ▶ We expect $\text{Cov}(X, Y)$ to be negative

Example: Computing covariance

	$X = -1$	$X = 0$	$X = 1$	f_Y	
$Y = -1$	$1/15$	$2/15$	$2/15$	$1/3$	$E[Y] = 0$ $E[X] = 0$
$Y = 0$	$2/15$	$1/15$	$2/15$	$1/3$	
$Y = 1$	$2/15$	$2/15$	$1/15$	$1/3$	
f_X	$1/3$	$1/3$	$1/3$		

$$\begin{aligned}
 \text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] = (-1)(-1) \cdot \frac{1}{15} + (-1)(1) \cdot \frac{2}{15} \\
 &\quad + (-1)(1) \cdot \frac{2}{15} + (1)(1) \cdot \frac{1}{15} \\
 &= -\frac{2}{15}
 \end{aligned}$$

Properties

① $\text{Cov}(X, X) = \text{Var}(X)$

► Proof: $\text{Cov}(X, X) = E[(X - E[X])(X - E[X])]$

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③ Covariance is symmetric, $\text{Cov}(X, Y) = \text{Cov}(Y, X)$

④ Covariance is a “linear” quantity

① $\text{Cov}(X, aY + bZ) = a\text{Cov}(X, Y) + b\text{Cov}(X, Z)$

② $\text{Cov}(aX + bY, Z) = a\text{Cov}(X, Z) + b\text{Cov}(Y, Z)$

Covariance and independence

- ① If X and Y are independent, X and Y are uncorrelated, i.e. $\text{Cov}(X, Y) = 0$
 - ▶ Proof: $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$

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 - Proof: $\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$
- ② If X and Y are uncorrelated, they may be dependent.

Example:

Dependent, but uncorrelated

	$X = -1$	$X = 0$	$X = 1$	f_Y
$Y = 0$	$1/8$	$1/4$	$1/8$	$1/2$
$Y = 1$	$1/4$	0	$1/4$	$1/2$
f_X	$3/8$	$1/4$	$3/8$	

$E[Y] = 1/2$
 $E[X] = 0$
 $E[XY] = 0$
 $E[XY] = E[X]E[Y]$

Correlation coefficient

Definition (Correlation coefficient)

The correlation coefficient or simply correlation of two random variables X and Y , denoted $\rho(X, Y)$, is defined as

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\text{SD}(X)\text{SD}(Y)}.$$

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- Result: $-\text{SD}(X)\text{SD}(Y) \leq \text{Cov}(X, Y) \leq \text{SD}(X)\text{SD}(Y)$
 - ▶ Simplify $E \left[\left(\frac{X-E[X]}{\text{SD}(X)} + \frac{Y-E[Y]}{\text{SD}(Y)} \right)^2 \right] \geq 0$ to get lower bound
 - ▶ Simplify $E \left[\left(\frac{X-E[X]}{\text{SD}(X)} - \frac{Y-E[Y]}{\text{SD}(Y)} \right)^2 \right] \geq 0$ to get upper bound

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- Using the above, $-1 \leq \rho(X, Y) \leq 1$

Correlation and random variables

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④ $|\rho(X, Y)|$ is close to one

- ▶ X and Y are strongly correlated
- ▶ Increase in X is likely to match up with an increase in Y

Problem

Consider the following joint PMF, where $-1/4 \leq x \leq 1/4$.

	$X = 0$	$X = 1$	f_Y
$Y = 0$	$1/4 - x$	$1/4 + x$	$1/2$
$Y = 1$	$1/4 + x$	$1/4 - x$	$1/2$
f_X	$1/2$	$1/2$	

$E[Y] = 1/2$
 $E[Y^2] = 1/2$
 $E[XY] = \frac{1}{4} - x$
 $E[X] = 1/2, E[X^2] = 1/2$

$$\text{Cov}(X, Y) = \frac{1}{4} - x - \frac{1}{2} \cdot \frac{1}{2} = -x$$

$$\text{Var}(X) = \frac{1}{4}, \text{Var}(Y) = \frac{1}{4}$$

$$\rho(X, Y) = \frac{-x}{\frac{1}{2} \cdot \frac{1}{2}} = -4x$$

Problem

Consider the following joint PMF, where $-1/4 \leq x \leq 1/4$.

	$X = c$	$X = c + 1$	f_Y
$Y = d$	$1/4 - x$	$1/4 + x$	$1/2$
$Y = d + 1$	$1/4 + x$	$1/4 - x$	$1/2$
f_X	$1/2$	$1/2$	

$$E[Y] = d + 1/2$$

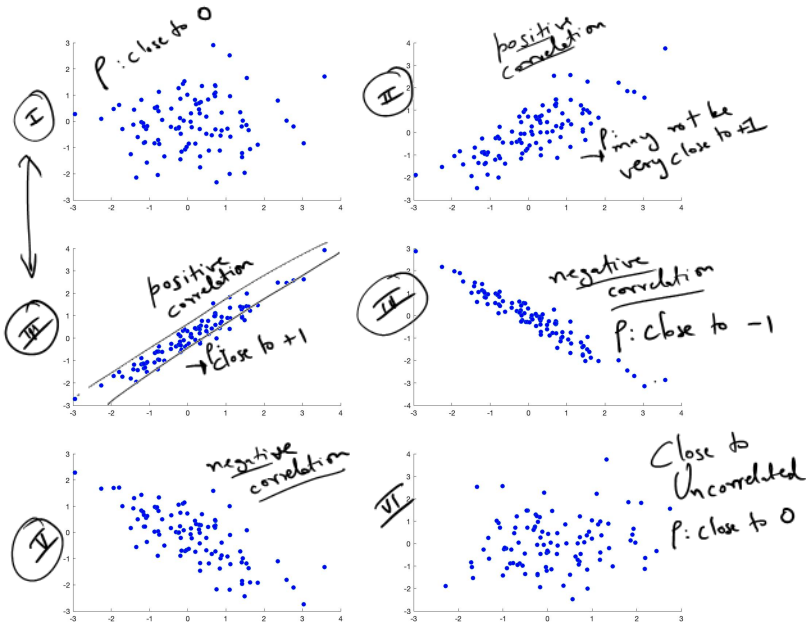
$$E[X] = c + 1/2$$

$$\begin{aligned}
 E[XY] &= cd\left(\frac{1}{4} - x\right) + c(d+1)\left(\frac{1}{4} + x\right) + (c+1)d\left(\frac{1}{4} + x\right) + (c+1)(d+1)\left(\frac{1}{4} - x\right) \\
 &= cd + \frac{c}{2} + \frac{d}{2} + \frac{1}{4} - x
 \end{aligned}$$

$$\text{Cov}(X, Y) = -x$$

$$\rho(X, Y) = -4x$$

Scatter plots and correlation



Subsection 5

Bounds on probabilities using mean and variance

Notation for mean and variance

- If there is no confusion about the random variable X ,
 - ▶ μ will denote the mean $E[X]$
 - ▶ σ^2 will denote the variance $\text{Var}(X)$
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 - ▶ μ will denote the mean $E[X]$
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 - ▶ σ will denote the standard deviation $\text{SD}(X)$
- If there are multiple random variables under discussion,
 - ▶ μ_X will denote the mean $E[X]$
 - ▶ σ_X^2 will denote the variance $\text{Var}(X)$
 - ▶ σ_X will denote the standard deviation $\text{SD}(X)$

What does mean say about the distribution?

Suppose the average marks in a course is 50/100.

- What fraction of students will have marks ≥ 50 ?

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 - ▶ It cannot be 1. Why?
 - ▶ Can it be 0.9? No.

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 - ▶ It cannot be 1. Why?
 - ▶ Can it be 0.9?

Average says something about the distribution of marks!

Standard units in statistics

- Consider a random variable X with mean μ and variance σ^2 .
- In an experiment, X may take a value that is close to μ or away from μ .
- $X - \mu$: measures the distance of X from the mean μ .
 - ▶ Could be positive or negative
- Standard units: The number of standard deviations that a realization of a random variable is away from the mean.
 - ▶ We expect $X - \mu$ to fall between $-c\sigma$ and $c\sigma$ for a small value of c
 - ▶ In other words, we expect X to fall between $\mu - c\sigma$ and $\mu + c\sigma$



A handwritten expression in black ink, circled, representing the formula for standard units: $\frac{X - \mu}{\sigma}$.

Examples

- Throw a pair of die, $X = \text{sum of the two numbers}$
 - ▶ $\mu = 7, \sigma \approx 2.42$
 - ▶ $P(|X - \mu| \leq \sigma) = P(4.58 \leq X \leq 9.42) = P(X \in \{5, 6, 7, 8, 9\}) = 2/3$
 - ★ So, $P(|X - \mu| > \sigma) = 1/3$
 - ▶ $P(|X - \mu| > 2\sigma) = P(X \in \{2, 12\}) = 2/36 \approx 0.056$

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 - ★ So, $P(|X - \mu| > \sigma) = 1/3$
 - ▶ $P(|X - \mu| > 2\sigma) = P(X \in \{2, 12\}) = 2/36 \approx 0.056$
- $X \sim \text{Uniform}\{1, 2, \dots, 100\}$
 - ▶ $\mu = 50.5, \sigma \approx 28.9$
 - ▶ $P(|X - \mu| > \sigma) = 58/100$
 - ▶ $P(|X - \mu| > 2\sigma) = 0$

Markov's inequality

Theorem (Markov's inequality)

Let X be a discrete random variable taking non-negative values with a finite mean μ . Then,

$$P(X \geq c) \leq \frac{\mu}{c}.$$

$$c = 100\mu$$
$$P(X \geq 100\mu) \leq \frac{1}{100} = 0.01$$

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Proof:

- $\mu = \sum_{t \in T_X} t P(X = t) = \sum_{t < c} t P(X = t) + \sum_{t \geq c} t P(X = t)$

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- Since the first sum is non-negative,

$$\mu \geq \sum_{t \geq c} t P(X = t) \geq \sum_{t \geq c} c P(X = t) = c \sum_{t \geq c} P(X = t) = cP(X \geq c)$$

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Other forms:

Non-negative *Markov*

- $P(|X - \mu| \geq \underline{c}) \leq \frac{\sigma^2}{\underline{c}^2}, P(\underline{(X - \mu)^2} \geq \underline{k^2 \sigma^2}) \leq \frac{1}{k^2}$

- $P(\underline{\mu - k\sigma} < \underline{X} < \underline{\mu + k\sigma}) \geq 1 - \frac{1}{k^2}$ ← complement of $|X - \mu| \geq k\sigma$

- $P(X \geq \mu + k\sigma) + P(X \leq \mu - k\sigma) \leq \frac{1}{k^2}$ $|X - \mu| \geq k\sigma \rightarrow \begin{matrix} X \geq \mu + k\sigma \\ \text{or} \\ X \leq \mu - k\sigma \end{matrix}$

- ▶ $P(X \geq \mu + k\sigma) \leq \frac{1}{k^2}, P(X \leq \mu - k\sigma) \leq \frac{1}{k^2}$

Compare actual and Chebyshev

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① $X \sim \text{Binomial}(10, 1/2)$, $\mu = 5$, $\sigma = \sqrt{2.5} \approx 1.58$

$$P(|X - 5| \geq 2\sigma) = P(X \in \{0, 1, 9, 10\}) \approx 0.021 \leq \frac{1}{4}$$

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$$P(|X - 5| \geq 2\sigma) = P(X \in \{0, 1, 9, 10\}) \approx 0.021$$

② $X \sim \text{Geometric}(1/4)$, $\mu = 4$, $\sigma \approx 3.46$

$$P(|X - 4| \geq 2\sigma) = P(X \in \{11, 12, \dots\}) \approx 0.056 < \frac{1}{4}$$

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- There are a lot of important theoretical implications as well. We will see these later on.